

may not match the requirements. And as the data resides with the cloud service provider, adhering to security policies becomes important. In such scenarios CASB comes to the rescue, fills in the gaps and complements cloud security services. This broker is placed between the cloud consumer and the cloud provider.

CASB provides security software as a service, addressing cloud service risks, enforcing the security policy, threat protection, data security, and compliance with regulations. It can be implemented as software running either locally or in the cloud, and has four primary characteristics.

- **Visibility:** Visibility of all the processes running on the cloud and ensuring they are authorised; validation and avoiding misconfigurations.
- **Compliance:** Following the organisation's own or mandated policies, like HIPAA and PCI.
- **Threat prevention:** Allowing only authenticated and authorised users to the granular level, including multi-level approvals or multi-factor authentication.
- **Data security:** Protecting and encrypting sensitive data at rest and in transit. Ensuring APIs are secure.

CASB, along with next-generation secure-web gateways (SWGs), monitoring and management of web APIs, and user/entity behaviour analytics (UEBA), provides the capability to deliver static and dynamic access to the management.

CASB is evolving into a huge service known as secure access service edge (SASE) architecture. SASE combines multiple security and networking technologies to provide comprehensive web and cloud security. Security service edge (SSE) is the convergence of multiple cloud-based security services as part of a SASE architecture.

Security awareness

'Structural awareness' solutions help manage risk, and protect systems and data.

- **Cloud compliance and best practices:** Visualisation and continual assessment of the environment, including the ability to enforce known good standards.
- **Instance and container visibility:** Monitoring and protection of containers throughout their life cycle.
- **Virtual private network:** Secure access to applications running in a VPC for remote employees.
- **Secure data:** Encryption and key management solutions to protect data and meet regulatory compliance standards.
- **Vulnerability assessment and management:** Visibility into the attack surface to discover and resolve security issues.
- **Software composition analysis:** Management of and security for open source licences.

- **Operational intelligence:** Aggregation and analysis of data for security, performance, and availability.
- **DevSecOps:** Automated and continuous process management for continuous integration and delivery of secure applications.
- **Cyber risk management:** Prioritised insight into the threats, vulnerabilities, and impacts of cloud assets.
- **Container security:** Minimal or hardened operating system that is specifically designed to run containers. Running similar security demand services on the same host limits the scope of intrusion.
- **Backup:** Maintaining backups ensures availability and helps to follow legal requirements such as holding information for seven years.
- **Permissions:** Authentication and authorisation at a granular level, and constant audits. 'Situational awareness' solutions detect security events and are capable of responding, recovering data, and enabling continual improvement.
- **Firewalls and proxies:** Provide fine-grained inspection of traffic for potential threats at both the network and application levels.
- **Endpoint detection and response:** Protects endpoints, including cloud workloads from zero-day and other attacks.
- **Intrusion detection systems:** Monitor networks and workloads for security events.
- **Backup and restoration:** Protects data from errors, failures, and accidental deletion.
- **Disaster recovery:** Provides additional flexibility to quickly recover from a disaster, ensuring that cloud workloads are available.
- **Security information and event management:** Ingests, correlates, and prioritises events for deeper visibility into anomalous behaviour and threat mitigation.
- **Workload isolation:** Provides dynamic security control for microservices, containers, and other workloads. 

References

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The author works in a leading bank as AVP. She is a perennial learner and loves the quote: "The more I learn, the more I realise how much I don't know." - *Albert Einstein*